Secure Micro Finance and Welfare Management platform using Blockchain

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Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

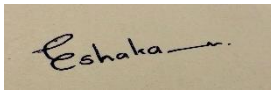
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Access to financial assistance and Micro loan is essential to helping low -income communities and improve economic stability. However , Many Microfinance institutions and Welfare programs struggle to evaluating the repayment ability of Loan Applicants . Traditional Loan Applicant Assessment methods often depend on manual review and limited financial data, which can result in inaccurate risk evaluations, higher default rates ,lengthy loan approval process and Loan Approval delays. These shortcomings emphasize the need for a intelligent ,data-driven decision making system. While several financial systems offer basic loan management features many current solutions do not have predictive capabilities to analyze customer characteristics and estimate probability of repayment .This lack of predictive analytics creates gap in research on the effective use of machine learning techniques to support loan approval decision in community focused financial system .To address this problem, this study introduces a loan and repayment prediction model that uses machine learning algorithms to assess the likelihood of loan repayment. The proposed solution analyzes various attributes such as Applicant income , expenses , Assets , Employment status , Share Amount ,Requested Loan Amount , Savings account balance, Existing loans, Repayment history to generate a repayment probability score and risk classification. Algorithms such as Logistic Regression ,Random forest and decision trees will be evaluated to determine the Loan repayment predictive model. Random forest will be used as the primary model due to its ability to capture financial attributes relationship and reduce overfitting. The model will be trained and evaluate to identify the most accurate prediction for assessing repayment capability and risk.This system will be implemented using technologies like Python for developing the machine learning model, data processing libraries such as Pandas and Scikit – learn and integrated model provide real-time loan risk predictions when applicants submit Loan request. The expected result is dependable predictive model that enhances the accuracy of loan approval decision and reducing financial risk. The Proposed system offers significant social and commercial advantages by promoting responsible Micro - lending practices, improving financial access for underserved rural communities, and helping microfinance organization make better informed lending decisions

Key words – Machine Learning , Loan Prediction, Repayment Risk analysis, Microfinance system, Predictive model

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TABLE OF CONTENT

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES ..... **Error! Bookmark not defined.**

LIST OF FIGURES ..... **Error! Bookmark not defined.**

LIST OF ABBREVIATIONS ..... **Error! Bookmark not defined.**

LIST OF APPENDICES ..... **Error! Bookmark not defined.**

1. INTRODUCTION 7

1.1 Background and Context 9

1.2 Literature Review 11

1.3 Research Gap 13

2. OBJECTIVE 15

2.1 Main Objective 15

2.2 Specific Objectives 15

3. METHODOLOGY 17

3.1 Research Approach 17

3.2 Technical Methods 17

3.3 Tools and Platforms 18

3.4 Data Collection and Ethical considerations 18

3.5 Validation Strategy and Performance metrics 19

4. HIGH-LEVEL SYSTEM ARCHITECTURE 20

4.1 Overview of Proposed Solution 20

4.2 High level System Diagram 20

4.2 Key Subcomponents 21

USER REQUIREMENTS 21

Functional Requirements 21

Non -Functional Requirements 22

Target market and customer persona 22

Value proposition 23

Cost estimation 23

Competitive advantage 24

7. BUDGET AND JUSTIFICATION 24

8. WORK BREAKDOWN STRUCTURE (WBS) 25

9. GANTT CHART 26

10. REFERENCES LIST 27

11. APPENDICES 28

LIST OF FIGURES

Figure 1 high level system diagam 20
Figure 2 work breakdown structure 25
Figure 3 Gantt Chart 26

LIST OF TABLES

Table 1Research Papers Comparison 14
Table 2Budget 24

LIST OF ABBREVIATIONS

Abbreviation	Full form
ML	Machine Learning
AI	Artificial Intelligence
RF	Random Forest
SPSS	Statistical Package for the Social Sciences
DSS	Decision Support System
UI	User Interface
DB	Database
F1 Score	Harmonic Mean of Precision and Recall

1. INTRODUCTION

Access of the financial assistance is an important factor of enhancing the economic stability of individuals and especially in rural areas and low income areas . Microfinance institutions and welfare organization such as Samupkara Gramiya Sanwaardhana Bank and Smurdhi provide financial solutions by way of Micro loans to enable individuals to offer funds to households , improve their living standards as well as , start businesses. Such financial services are relevant in ensuring alleviation of poverty , economic growth ,encouraging entrepreneurship within underserved community and ease day to day life. However ,this is also one of the major challenge facing such institution which require assessing the ability of an applicant to repay a Loan.

In most developing Regions such as south asia , the process of loan evaluation remain mostly manual and depend heavily on review of financial officers . Income records, employment status and household financial information are some of the requirements that loan officers usually examine when evaluating the applicant . However , this procedure is usually time consuming and failure to provide a clear representation of the financial ability of the applicant .Also Other issues like patterned income ,informal incomes and weak credit records complicated the decision making process. This will lead to lenders issuing loan to borrowers who cannot repay or refusing loans to individuals who can repay.

Loan defaults are one of the major financial risk of leading institutions. When borrowers fail to repay loans at the on time ,institution experience financial losses and this may affect with their capacity to offer financial support in future . At the same time, borrowers who are not financially viable to get loans might end up in long term debt traps. Therefore , accuracy in the evaluation of loan and prediction of loan repayment is the key in the processes of the institutional sustainability as well as responsible lending practices.

Many of the traditional micro loan management systems that are practiced by traditional institution in sri lanka are mainly aimed at keeping records of the borrowers and monitoring the repayment schedules manually . Although these systems can organize financial information, in most cases they do not have intelligent decision support system capable of analyzing financial patterns and predict repayment outcomes. Traditional credit scoring systems are usually developed to suit formal banking industries and do not necessarily reflect

the financial practices of the individuals in micro finance environment where income earning and financial practices are more diverse.

The latest advances in the fields of the data analytics and machine learning create new possibilities to improve the financial decision making systems. With data on historical borrowers and financial characteristics, predictive model are able to identify patterns in terms of loan repayment behavior. The machine learning algorithms will be able to analyze multiple variables that include Applicant income , expenses , Assets , Employment status , Share Amount ,Requested Loan Amount , Savings account balance, Existing loans, Repayment history and then give estimate of whether the loan will be repaid successfully. These prediction abilities enable financial institutions to make an informed decision and minimize risk lending.

This research propose Loan and Repayment Prediction Model ,which involves the use of machine learning in order to assess the financial information of applicant and provide insights to predict the loan evaluation. The system will set out to identify the ability of an applicant to afford a loan ,estimate potential repayment duration, risk level of borrower and recommend suitable Loan type/option based on financial capacity. The proposed system will be beneficial to sustainable financial practices as it aims to enhance the speed, accuracy and fairness of lending decision by incorporating predictive analytics into the process of approving loans.

The implementation of loan prediction can generate a large social and economics advantages. Better loan screening systems can assist financial institutions to reduce the risks of defaulting and at the same time, provide the deserving candidates with the relevant financial capacity. Moreover, these systems can be used to help achieve larger objectives of financial inclusion and economic empowerment, as more people are able to receive responsible and well-organized financial assistance.

1.1 Background and Context

Affordable Financial services are the basic requirement of economic development and social stability. Community welfare programs , Cooperative communities and Microfinance institution such as Samupakara Gramiya Sanwardhana Bankhave crucial role of offering financial support to people who might not be eligible to receive the services from traditional banking system. These banks offer affordable Micro – Loans to help people to manage household, start businesses, Invest in agriculture ,Construction support ,educational and healthcare expenses. However ,the identification of the financial capacity of the loan applicant to repay a loan is a complicated and critical task.

The issue is mostly experienced in the rural and low-income communities whereby formal financial statements are minimal and the sources of income ,expenses and personal information might be inconsistent. Loan officers have been using manual methods of evaluation which includes going through financial records, interviewing applicant and guarantors , and passing judgement based on scanty information and also investigating within the household and neighborhood. This strategy may result in inaccurate loan decisions, whereby a person who capable to repay their loans may be denied an opportunity, while another applicant who is not stable financially may be receive a loan that he cannot repay. Such inaccurate credits can raise the financial risk of the lending institutions as well as applicant and introduce financial burdens that may be long term to the Loan Applicant.

The need to enhance a loan evaluation system is crucial in the global finance trends. As per reports by the World Bank , it is estimated that 1.4 billion adults in the entire world are not banked and they depend on informal or microfinance lending systems one way or the other. Also in sri lanka around 2.9 Million peoples are active microfinance borrowers ,most of them are from rural low income community. Meanwhile, the financial institutions still have difficulties when it comes to loan default risks and poor credit evaluation process . When there are no adequate mechanisms of assessing the borrowers, in most developing nations, default rates in microfinance lending may be quite high. These issues shine light on the necessity of more credible and evidence-based methods of assessing loan applications and forecasting their repayment behavior

The current digital loan management tools are mainly concerned with the preservation of the records of the borrowers, monitoring of the repayment plans, and the generating financial documents. As much as these systems enhance the efficiency of operations, they do not usually have intelligent decision-support functions which have the ability to analyze the data on borrowers and determine the possibilities of them repaying. The Traditional credit score systems are normally design to a formal banking setting and use structured credit histories that most microfinance borrowers lack. Consequently, such techniques might not capture the financial characteristics of those whose levels of income are not regular or who work informally.

In order to overcome these limitations, the proposed research presents a Loan and Repayment Prediction Model which incorporates machine learning in the loan evaluation process. The suggested system examines various financial and behavioral characteristics of the applicants such as Applicant income , expenses , Assets , Employment status , Share Amount ,Requested Loan Amount , Savings account balance, Existing loans, Repayment history . Using predictive analytics, the model is able to determine the probability of repayment, categorizing the level of risk associated with a specific borrower, the appropriate repayment period and the option of loan type that the applicant can reasonably repay. This has been done in order to help financial institutions as well as borrower to make more precise, transparent and data-driven lending decisions.

The intelligent financial decision-support systems are also developed towards the wider development objectives of the developing region. The study is in line with United Nations sustainable development goal number 8 that advocates sustainable economic growth, productive employment as well as access to financial services. The proposed system will enhance responsible lending practices by streamlining and making loan approval more equitable to underserved communities and increasing financial accessibility. Moreover, the study belongs to Data Science and Artificial Intelligence research cluster as it is aimed at using machine learning algorithms and predictive analytics methods to process financial data and create actionable information. Introduction of intelligent predictive models in the financial systems is a significant move of coming up with smarter and sustainable financial technologies capable of fostering economic growth at both local and global scales.

1.2 Literature Review

Financial inclusion and credit access have gained relevance in the economics development studies, especially in the developing nations where the majority of people depend on the microfinance institutions to acquire their financial resources. The microfinance institutions are small scale lending institutions and sources of financial services to the low income communities who do not necessarily have access to the normal banking institutions.According to **"Gender and Other Repayment Determinants of Microfinancing in Indonesia and Sri Lanka"** conducted by **Gemunu Nanayakkara** and **Jenny Stewart**. The study aimed to develop models to predict repayment success in microfinance loans using logistic regression analysis. The researchers analyzed 1,109 loan records collected from microfinance institutions in Indonesia and Sri Lanka. The findings revealed that several factors such as gender, borrower age, loan cycle, loan purpose, and loan officer visiting frequency significantly influence repayment performance in Sri Lanka. The developed models achieved more than 70% prediction accuracy in identifying repayment success. However, the study mainly focused on statistical analysis of repayment determinants and did not propose an automated decision-support system for predicting loan affordability or recommending suitable loan plans for borrowers.

Several studies conducted in Sri Lanka have further explored the factors influencing loan default and repayment behavior among microfinance borrowers.**D. T. Priyankara** and **E. A. G. Sumanasiri** conducted research on **"Determinants of Microfinance Loan Default: An Empirical Investigation in Sri Lanka"** published in the *South Asian Journal of Social Sciences and Economics* in 2019. The objective of this study was to identify factors influencing loan default among microfinance borrowers in Sri Lanka. The researchers used a quantitative research approach and collected data from 133 microfinance borrowers using structured questionnaires. Statistical analysis was conducted using SPSS software to identify relationships between borrower characteristics and loan repayment outcomes. The results indicated that borrower family characteristics, loan group dynamics, and macroeconomic factors significantly influence loan default risk. While the study provides valuable insights into factors affecting loan default, it mainly focuses on statistical analysis and does not implement predictive machine learning models for automated loan risk evaluation.

Research related to microfinance loan repayment and risk management has been widely conducted in the Sri Lankan context **.D. H. Wickramasinghe** conducted research on **"Study on Factors Influencing Microfinance Loan Repayment"** with special reference to Samurdhi Banks in Sri Lanka. This research examined the variables that influence the repayment behavior of microfinance borrowers by analyzing borrower characteristics, business characteristics, and loan characteristics. Data was collected from default borrowers and analyzed using statistical tools. The findings revealed that borrower behavior, business performance, and loan-related factors significantly affect repayment performance. Although the research identifies key determinants of repayment, the study is limited to descriptive statistical analysis and does not develop predictive models that can assist financial institutions in automated loan approval or repayment forecasting. Similarly **E. A. G. Sumanasiri** and **D. T. Priyankar**a conducted another study titled **"Microfinance Loan Default in Sri Lanka: A Case Study of Matugama Division"** which examined factors contributing to loan default within microfinance institutions in Sri Lanka. The researchers employed survey-based data collection and statistical analysis techniques to evaluate borrower characteristics and institutional factors affecting loan default. The results highlighted the importance of borrower financial behavior, credit history, and institutional control mechanisms in managing loan defaults. However, the study mainly focuses on analyzing the causes of loan default rather than developing predictive models capable of evaluating borrower risk during the loan approval stage.

In recent years, machine learning techniques have been increasingly applied to financial risk prediction and credit scoring. Machine learning algorithms can analyze large volumes of financial data and identify patterns that may not be visible through traditional statistical methods. For example, Researched conducted by **H A P L J Hettiarachchi** and Ruwan **Punchi manage** conducted research on **"Analysis of Loan Delinquency Prediction Based on Multinomial Logistic Regression and Random Forest"** published in 2025. This study investigated predictive modeling approaches to classify loan delinquency levels using a large dataset containing 43,644 loan records from a Sri Lankan bank. The researchers applied multinomial logistic regression and **Random Forest** models to categorize loan performance into multiple risk levels ranging from performing loans to severely delinquent loans. The results showed that Random Forest achieved higher predictive accuracy with a low out-of-bag error rate, while logistic regression provided detailed insights into how variables such as interest rate and loan age influence loan delinquency risk. However, the study mainly focuses

on loan delinquency classification and does not include borrower affordability analysis or personalized loan recommendation mechanisms.

1.3 Research Gap

Most of the studies on the analysis of microfinance loans that exist are aimed at identifying determinants that help in measuring loan repayment performance or forecasting loan default by using statistical methods. Other studies like the work of Nanayakkara and Stewart also applied logistic regression models in the determination of repayment determinants including borrower demographics, the purpose of the loan, and the number of visits to the borrower by the loan officer. On the same note, research carried out in Sri Lanka has investigated the factors that determine the loan default and repayment behavior through statistical analysis survey-based and SPSS . Although these studies offer a good understanding of borrower behavior, and determinants of repayment, they are mainly aimed at identification of behaviors between variables and not coming up with automated predictive models that aid in real-time decision making.

There are some more recent works that tried to use machine learning to perform financial prediction. Thus, in 2022, a study at the International Conference on Science and Technology offered a solution to predict interest rates on microfinance with the help of regression models and machine learning in processing loan applications by handwriting recognition . Even though the research shows how machine learning can impact the microfinance sector, it is most concerned with the aspects of interest rates prediction and processing documents as opposed to verifying loan applicant capacity or ability to repay a loan. Moreover, machine learning prediction models of loan delinquency like Rand Forest and logistic regression have also demonstrated their potential in loan default risk prediction . Nevertheless, the majority of available predictive models are not exhaustive and focused on binary risk (e.g., default vs. non-default) or categorical risk (i.e., good vs. bad) classification of loans without the complete provision of decision support functionality (e.g., the identification of the appropriate type of loan, the estimation of the loan repayment period, or the prescribing of the optimal loan repayment schedule to the borrowers). Hence, there is a serious research gap in coming up with a cohesive intelligent system that is not only able to forecast and predict the risk of loan issuance, but is also able to consider the affordability of the borrower, recommend the right

type of loan, calculate the duration of repayment and recommend the best repayment schedules. This system would be more practical in decision support to microfinance institutions in the process of loan appraisal. The gap filled by the proposed research is the creation of a machine learning-driven loan and repayment prediction model that will analyse the financial characteristics of the borrowers, including income, expenses, assets, employment status, existing loans, and repayment history to estimate the affordability of a loan and risk levels. The system will help microfinance institutions to make better lending decisions and minimize loan default risks by combining predictive analytics with decision-support features.

Table 1Research Papers Comparison

Method/Feature	**Paper 01**	**Paper 02**	**Paper 03**	**Paper 04**	**Proposed system**
Statistical analysis	✓	✓	✓	✓	✓
Machine learning prediction model	✓	✗	✗	✓	✓
Loan default/delinquency analysis	✓	✓	✓	✓	✓
Financial behavior analysis	✓	✓	✓	✓	✓
Loan Affordability evaluation	✗	✗	✗	✗	✓
Automated loan decision support	✗	✗	✗	✗	✓
Loan recommendation/ repayment planning	✗	✗	✗	✗	✓

2. OBJECTIVE

2.1 Main Objective

The primary aim of the research is to create and build a machine learning-friendly loan and repayment prediction tool that is able to assess and determine the affordability of the borrowers, estimate the repayment and evaluate the riskiness of the loans taken by microfinance institutions. The system will also examine financial and demographic characteristics that will include; income, expenses, assets, employment status, existing loans, balance in savings and repayment history of a loan applicant to come up with an answer regarding whether the applicant can afford a loan requested. Besides this, the model will prescribe the appropriate types of loans, approximate the best repayment plans, and indicate the amount of financial risk to be taken on loan granting. Through the development of predictive analytics into the loan appraisal system, the proposed system will assist microfinance institutions to make better lending choices and minimise the risks of loan defaults and enhance financial sustainability.

2.2 Specific Objectives

Dataset Collection and Preparation

Collect and organize a dataset containing borrower financial and demographic attributes such as applicant income, expenses, assets, employment status, savings account balance, existing loans, repayment history, requested loan amount, and share amount. The collected dataset will be cleaned, normalized, and structured to ensure high data quality for machine learning model development.

Data Preprocessing and Feature Engineering

Perform data preprocessing techniques including handling missing values, normalization, and feature selection to improve the quality and relevance of the dataset. Important predictive features will be extracted and prepared to enhance the performance and reliability of the loan prediction model.

Machine Learning Model Development for Risk Prediction

Develop and train a supervised machine learning model to classify the risk level of loan applicants and predict the likelihood of successful repayment. Ensemble learning algorithms such as Random Forest will be explored due to their ability to handle complex financial datasets and improve prediction accuracy.

Loan Affordability and Repayment Duration Prediction

Design predictive mechanisms to estimate whether the applicant can afford the requested loan and determine the appropriate repayment duration based on financial capacity. The model will analyze borrower income, expenses, and liabilities to recommend feasible repayment schedules.

Loan Recommendation and Decision Support

Develop a decision-support mechanism that recommends suitable loan types and repayment plans for borrowers who cannot afford the originally requested loan amount. This component aims to assist financial institutions in offering alternative loan options while maintaining manageable financial risk levels.

System Integration and Performance Evaluation

Integrate the developed machine learning model into a system that supports real-time loan analysis and decision-making. The performance of the model will be evaluated using metrics such as prediction accuracy, precision, recall, and confusion matrix analysis to ensure reliability and effectiveness.

3. METHODOLOGY

3.1 Research Approach

The study will follow the design science research and an experimental approach to conduct research aimed at designing and testing a machine learning-based loan and repayment prediction model. The design science approach concentrates on the development of a technological artifact, in the present case, a predictive decision-support system, which will resolve real-world issues in the microfinance loan appraisal. Experimental component entails training and testing machine learning models with past data on borrowers in order to determine trends in terms of debt affordability, repaying abilities, and monetary risks. The suggested system will be able to examine the financial characteristics of the borrowers and will forecast, loan approval possibility, riskiness, loan termination period and recommended loan services. To test the reliability and predictive accuracy of the developed model, the effectiveness of the created model will be justified by means of experimental verification based on training and testing datasets.

3.2 Technical Methods

The suggested system will apply to supervised machine learning to examine financial data of borrowers and forecast future outcomes of loan repayment. Attributes in the dataset will include applicant income, expenses, assets, employment, savings account balance, requested loan amount, current loan, share amount and repayment history. The main algorithm that has been chosen to build the predictive model is the algorithm of random forest which is a technique of ensemble learning and it builds several decision trees and combines the results of these decision trees to enhance accuracy and minimise the issue of overfitting. Random Forest is appropriate in a scenario of financial prediction issues as it is able to deal with complicated associations involving more than two variables and works well with structured tabular data.

The development process include several stages :

- Data Preprocessing : handling missing values, normalization
- feature selection : Identify most significant variables about loan repayment ability
- Model Training : Training the Random Forest model using loan and repayment history

- Prediction Tasks :
    - Loan can afford or cannot afford
    - Loan risk level prediction(Low , medium , high risk)
    - Repayment duration
    - Loan type recommendation

3.3 Tools and Platforms

Several software tools and frameworks that help with the development and implementation of machine learning models will be used in the development of the system.

- Python Programming Language - Strong language for machine learning development and analysis
- Pandas – Used for data preprocessing, cleaning and data manipulation
- Scikit learn – Used to implementing machine learning algorithmwith Random forest, model training and evaluation
- NumPy – Support for numerical computations
- Matplotlib – Used to visualization of model performance

These tools are appropriate to develop predictive financial models because they offer effective machine learning capabilities, extensive documentation, and strong community support.

3.4 Data Collection and Ethical considerations

The historical loan and repayment data collected from microfinance loan records in **Samupakara Gramiya Sanwardha Bank** makes up the dataset used in this research. Financial details including applicant income, expenses, assets, employment status , savings account balance, requested loan amount, existing loan, share amount and repayment history are all included in the dataset.

Before the model is trained, data preprocessing techniques will be used to guarantee data consistency and quality. Sensitive personal data will be anonymised and managed securely to

preserve borrower privacy in accordance with ethical research principles. In order to ensure adherence to ethical norms and responsible data usage, the data will only be used for research and model development.

3.5 Validation Strategy and Performance metrics

The dataset will be split around 80:20 into training and testing sets in order for evaluating the effectiveness of the suggested machine learning model. The Random Forest model will be trained using the training dataset, and prediction accuracy will be evaluated using the testing dataset.

The effectiveness of the model will be evaluated using a number of performance metrics:

- Accuracy : Determine overall correctness of prediction
- Precision : Evaluate how many positive case are correct in prediction
- cc : Measure ability of model to identify positive cases
- F1-Score : Provide balance measure combining Precision and Recall
- Confusion Matrix : Visualize classification performance

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution

In order to help microfinance institution(Samupakara Gramiya Sanwardha Bank) evaluate loan applications and forecast repaying results, the suggested solution is an intelligent loan management platform. Data collecting, document processing, machine learning-based prediction, and secure financial record management are just a few of the technological elements that the system includes.

The architecture is made up of a number of integrated modules that collaborate to process borrower data and produce predicted insights. Initially, the user interface or digital loan application forms are used to gather borrower financial and personal data. Structured data is extracted from supplied documents by the document processing module. The predictive analytics module receives the processed data and assesses the borrower's capacity to repay the loan.

4.2 High level System Diagram

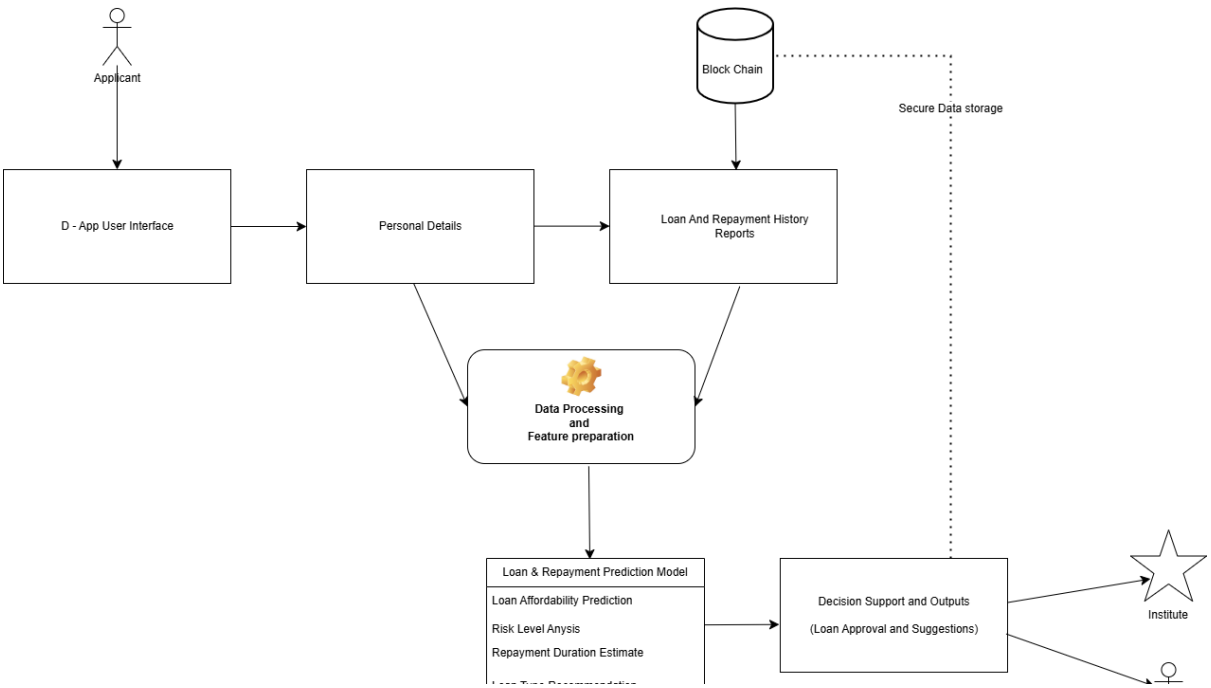


Figure 1 high level system diagam

4.2 Key Subcomponents

- **Data Preprocessing Module** – Clean and prepare dataset by handling missing values and normalization attributes
- **Feature Engineering -** Important predictive features such as income-expense ratio, debt-to-income ratio, and repayment history indicators are derived to improve model performance.
- **Machine Learning Prediction engine -** This module trains and executes the predictive model using the Random Forest algorithm to classify loan risk level and affordability
- **Decision Recommendation Module -** based on prediction results, this module recommends suitable loan types, repayment durations, and instalment schedules.

USER REQUIREMENTS

Functional Requirements

- Users will be able to submit loan application information, including financial and personal details, using the system
- Borrower financial data will be processed by the system and made ready for machine learning analysis
- The borrower's ability to pay the proposed loan amount will be predicted by the system.
- The system will use the borrower's financial capacity to predict the repayment period and installment schedule.
- The system will assess the degree of risk involved in giving the applicant a loan
- If the requested loan amount is too high, the system will provide other loan possibilitie
- The system will safely store loan application information and forecast outcomes in the system database.

Non -Functional Requirements

- Security - Sensitive borrower data must be protected by the system using secure authentication procedures and encryption.
- Performance - After receiving the input data, the prediction model should produce loan evaluation findings in a matter of seconds.
- Usability - Loan officers and applicants should be able to use the user interface without the need for sophisticated technical knowledge.
- Scalability - The system should be able to handle growing datasets and an increase in loan applications without experiencing any reduction in performance.
- Reliability - When handling loan prediction jobs, the system must continue to operate consistently and accurately.

5 COMMERCIALIZATION PLAN

Target market and customer persona

Microfinance institutions, cooperative banks, rural financial organisations, and small lending institutions such as Samupakara Gramiya Sanwardhana Bank , Samurdhi that offer loans to people with low incomes are the primary target market for the proposed system. These organisations frequently deal with a high volume of loan applications and have a difficult time determining borrower affordability and forecasting default risks.

A loan officer or financial analyst employed by a microfinance organization is the usual customer persona. These experts are in charge of assessing loan applications, establishing borrower eligibility, and reducing the likelihood of loan default. They need trustworthy instruments that can swiftly evaluate borrower financial information and produce precise risk evaluations.

Financial managers and lending institution decision-makers are examples of secondary users who need data-driven insights to enhance loan portfolio management and financial sustainability.

Value proposition

A data-driven decision support tool is offered by the proposed system to assist microfinance institutions in making more precise loan decisions. The system's primary value proposition consists of

- Automated loan affordability assessment based on borrower financial data
- Prediction of loan repayment capability and repayment duration
- Risk level analysis
- Recommendation of suitable loan options

The system can analyse datasets and find patterns that are challenging to find through manual review by utilising machine learning models like Random Forest. For lending institutions, this increases operational efficiency, lowers financial risk, and improves decision accuracy.

Cost estimation

The system's initial development and ongoing expenses consist of the following:

- Development cost
  - Development of machine learning models
  - System integration and software development
  - Design and testing of user interfaces

- Operational Cost
  - Database storage and cloud hosting
  - Updating and maintaining the system

- Costs of Data Management
  - Mechanisms for protecting privacy and security
  - Infrastructure for data backup and storage

Competitive advantage

When compared to current digital lending tools and conventional loan appraisal techniques, the suggested methodology offers a number of benefits.

- Predictive analysis powered by AI that increases loan risk assessment accuracy
- Loan affordability analysis, payback forecasting, and loan advice are examples of integrated decision support.
- Automated data processing reduces loan officers' manual labour
- decreased loan default rates to increase microfinance institutions' financial sustainability

The suggested system offers a complete loan decision-support architecture that incorporates several predictive functions into a single platform, in contrast to many current solutions that just carry out credit scoring or interest rate estimation.

7. BUDGET AND JUSTIFICATION

Table 2Budget

External Storage Device	Used for secure backup and storage of datasets and project files	LKR 20000
Cloud Hosting	Hosting the application and prediction model	LKR 10,000 / year
Cloud Storage	Secure storage of borrower datasets and model outputs	LKR 5,000 / year
System Testing and Documentation	Testing system functionality and preparing documentation	LKR 10,000
Transport		LKR 7000
Total estimate budget		LKR 52000

8. WORK BREAKDOWN STRUCTURE (WBS)

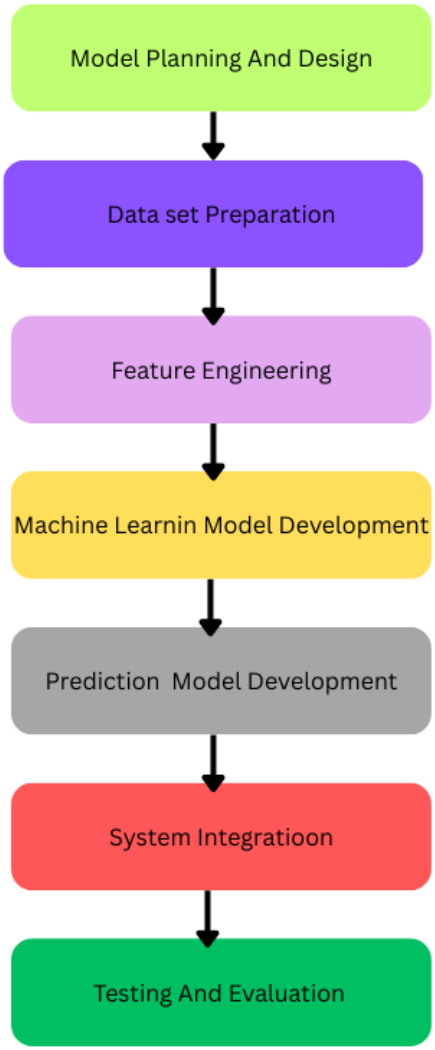


Figure 2 work breakdown structure

9. GANTT CHART

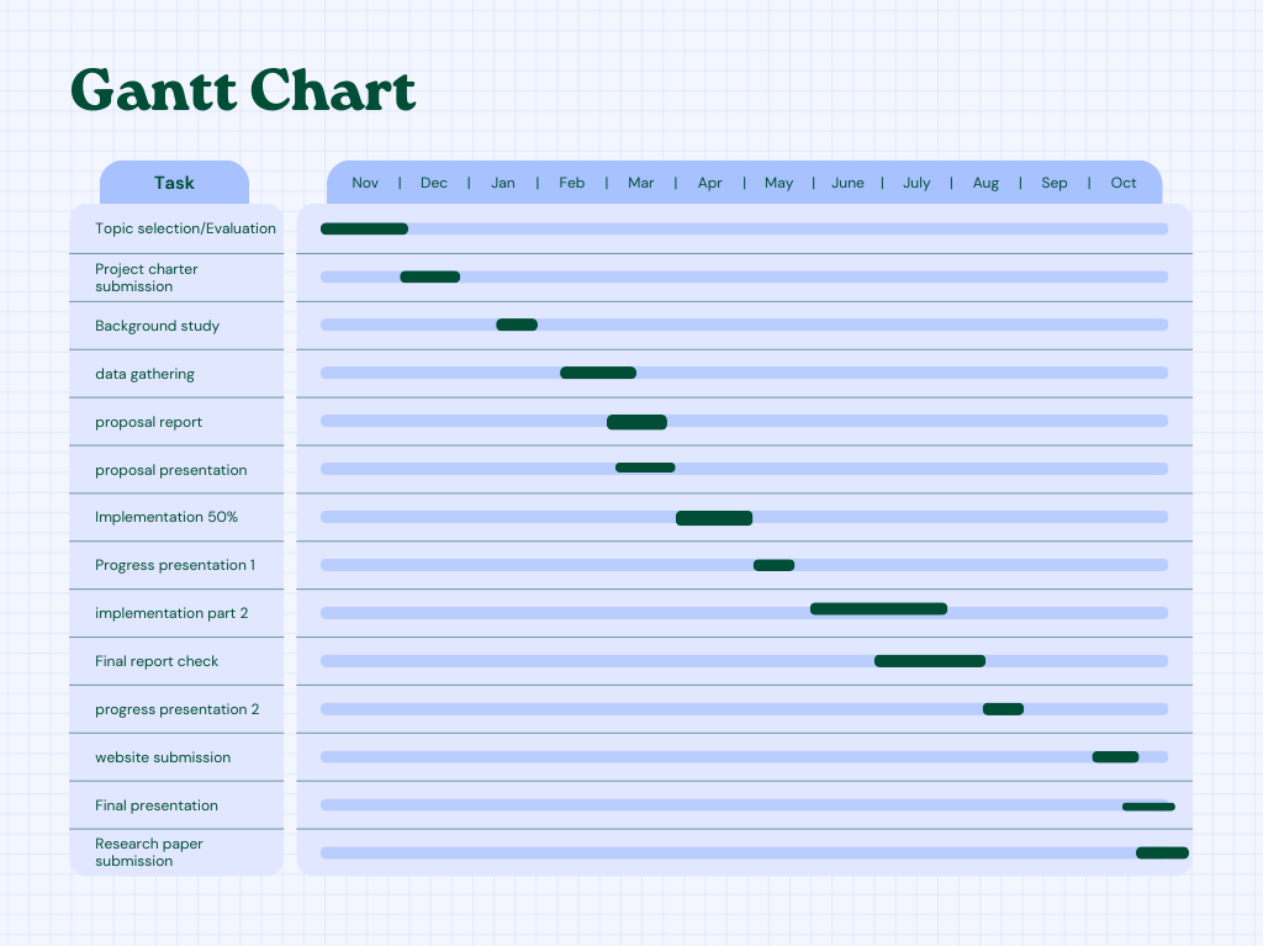


Figure 3 Gantt Chart

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11.APPENDICES

Appendix A – Loan Application Data Attributes (Provided by Samupakara Gramiya Sanwardhana Bank)

Sample Ledger Portfolio (Microfinance Credit Scoring)

Field	Details / Example	Purpose
Customer ID / Account No.	2025-MF-001	Unique identifier
Customer Name	John Silva	Identification
National ID / Passport No.	123456789V	Verification
Age / Gender	35 / Male	Demographics
Address / Location	Colombo, Sri Lanka	Regional mapping
Household Size	4	Dependency ratio
Loan Type	Gold-backed / Micro loan	Product category
Loan Amount	LKR 50,000	Exposure
Collateral Details	22K Gold chain, 25g	Security
Monthly Income	LKR 35,000	Repayment capacity
Monthly Expenses	LKR 20,000	Debt-to-income ratio
Savings / Deposits	LKR 5,000	Financial discipline
Existing Loans	LKR 15,000 (other MFI)	Over-indebtedness check
Repayment History	95% on-time	Reliability
Attendance (Group Meetings)	90%	Commitment
Mobile Money Usage	Active (weekly transactions)	Alternative data
Utility Bill Payments	On-time	Responsibility
Risk Flags	None	Fraud detection
Credit Score	78/100 (Band B)	Risk rating
Decision	Approved – Conditional	Loan eligibility
Officer Remarks	Customer reliable, collateral verified	Governance
Date of Entry	08 Dec 2025	Audit trail

✅ **Key Notes**

- This **ledger portfolio** can be digitized into your app database.
- Each row = one customer record.
- You can add **columns for audit logs** (who entered, who approved).

Appendix B – Bank Requirement Document

Consolidated Credit Scoring Framework (Microfinance)

1. Objectives

- Assess repayment capacity of low-income customers.
- Balance financial inclusion with risk control.
- Provide quick, transparent scoring for loan officers.

2. Data Collection Criteria

- **Customer demographics**: age, gender, household size, education.
- **Financial data**: income sources, monthly expenses, savings, existing loans, collateral (gold/assets).
- **Behavioral data**: repayment history, attendance at group meetings, transaction frequency.
- **Alternative data**: mobile money usage, utility bill payments, community references.

3. Data Processing Rules

- Standardize inputs (normalize income ranges, expense categories).
- Handle missing/incomplete data with fallback rules.
- Apply fraud detection checks (fake gold, duplicate accounts, ID mismatches).

4. Credit Scoring Model

- **Traditional methods**: logistic regression, decision trees.
- **Advanced methods**: machine learning (random forest, gradient boosting).
- **Hybrid scoring**: combines financial, behavioral, and alternative data.
- **Weighting example**:
  - Repayment history → 40%
  - Income stability → 30%
  - Alternative data → 20%
- **Medium risk** → smaller loan, higher monitoring.
- **Low risk** → full loan approval.
- Include early warning signals for fraud or default.

6. Operational Interface

- Simple tools for field officers to capture and review data.
- Offline capability with later synchronization.
- Multilingual support (Sinhala, Tamil, English).
- Display: customer profile, credit score, loan eligibility decision.

7. Integration

- Connect with core banking systems or microfinance MIS.
- Link to mobile money platforms for repayment tracking.
- Enable audit logs for governance and compliance.

8. Governance & Compliance

- Align with CBSL guidelines and microfinance regulations.
- Ensure customer data privacy and consent.
- Document scoring methodology for audit transparency.

9. Testing & Validation

- Pilot with small customer groups.
- Compare predicted scores vs. actual repayment behavior.
- Refine weights and thresholds.

10. Continuous Improvement

- Update scoring model with new repayment data.
- Add fraud detection modules (fake gold, identity mismatch).